



## System Design

Explain the advantages of using agile tools to support an agile development process.

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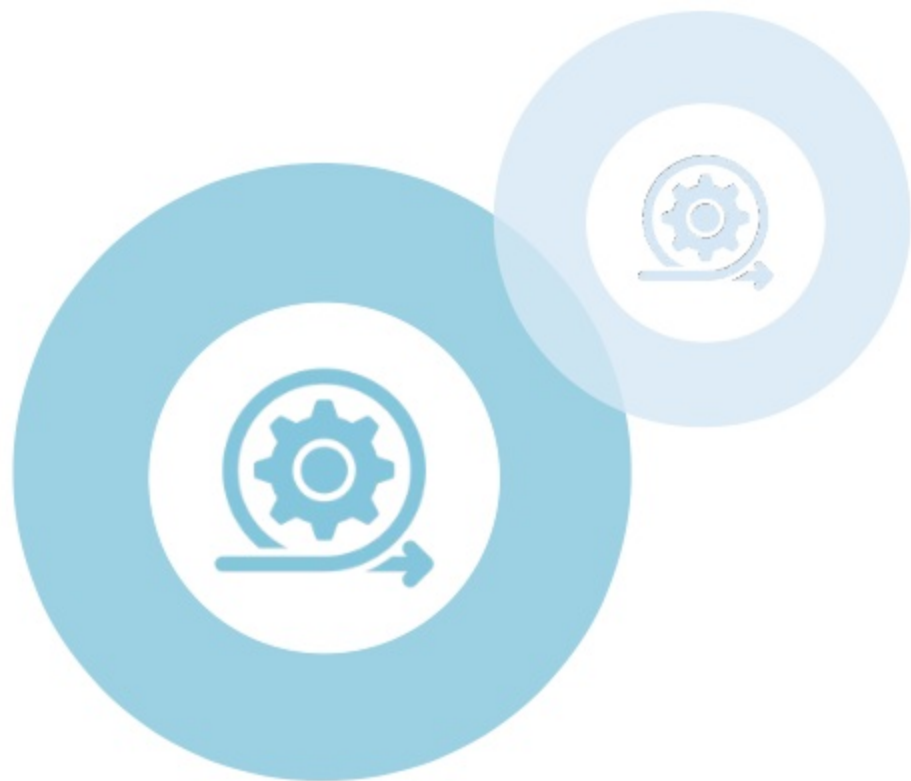
-  [Agile Development Tools](#)
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## After studying this topic, you should be able to:

- ✎ Identify the types of tools that can support an agile development process.
- ✎ Identify the features and advantages of Salesforce DX in the context of agile development.
- ✎ Identify the features and advantages of agile project management tools.
- ✎ Describe the benefits of using an agile tool for the given business requirements.



# Introduction

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In Salesforce development, leveraging agile tools like **Salesforce DX** and **agile project management tools** significantly enhances team productivity and project management. Salesforce DX streamlines the development lifecycle with features like **Salesforce CLI**, **scratch orgs**, and **Dev Hub**, supporting rapid iteration and quality assurance.

Agile project management tools, such as **Agile Accelerator**, facilitate **task management**, **sprint planning**, and **real-time collaboration**, allowing teams to adapt quickly and efficiently to changes. These tools are foundational in applying agile methodologies, ensuring teams can deliver superior Salesforce solutions swiftly and effectively.



# Advantages of Agile Tools

## Agile Development Tools

Tools such as Salesforce DX and agile project management tools, such as Agile Accelerator, can be utilized to support an agile development process.

## Agile Project Management Tools

Agile project management tools offer various features that support the iterative and incremental nature of an agile development process.

## Salesforce DX

Salesforce DX is a suite of tools designed to streamline the entire development lifecycle, improving collaboration, efficiency, and quality.

## Advantages of Agile Tools

Agile tools offer advantages such as improved visibility, enhanced collaboration, centralized repository, scalability, and predictability.



# Agile Development Tools

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# Agile Development Tools

In Salesforce development, two types of **agile tools** can be utilized to enhance **team productivity, collaboration, and project management**.

## SALESFORCE DX

Salesforce DX (Developer Experience) is a set of tools used to **streamline the entire development lifecycle**, improving collaboration, efficiency, and quality.



## PROJECT MANAGEMENT TOOLS

Agile project management tools enable teams to **collaborate more effectively, manage tasks efficiently**, and **deliver value faster** and with **better quality**.

# Salesforce DX for Agile Development

**Salesforce DX** is designed to support an **agile development process**. Its features enable faster delivery, higher quality, and more efficient collaboration.

## SALESFORCE CLI

Salesforce CLI allows developers to **create** and **manage** Salesforce environments, **synchronize source code**, **execute tests**, and **deploy applications**.

## SCRATCH ORGS

Scratch orgs are **temporary, configurable, and disposable** Salesforce environments that support features like **isolated development**, **automated testing**, and **continuous integration**.



## SOURCE-DRIVEN DEVELOPMENT

**Changes** can be managed and tracked in a **version control system (VCS)** before deploying to Salesforce environments, which **enhances collaboration** and **version control practices**.

## DEV HUB

Dev Hub is the main Salesforce org that supports the **collaborative and agile aspects** of development by **enabling the management of scratch orgs**.

# Advantages of Salesforce DX

**Salesforce DX** aligns well with agile development methodologies by providing tools that support **rapid iteration**, **collaboration**, and **quality assurance**.



Salesforce DX offers **enhanced developer productivity** through scratch orgs and Salesforce CLI. Developers can quickly spin up scratch orgs, **speeding up development and testing cycles**.



Source-driven development facilitates **better tracking of changes** and **collaboration** among team members, leading to **fewer conflicts** and **more efficient development processes**.



Scratch orgs allow for **rapid testing of new features** in a controlled environments, supporting agile practices like continuous integration and delivery.



The configurability offered by Dev Hub and the ability to create and dispose off scratch orgs allows teams to **adapt quickly to changing requirements** and **scale their development efforts**.

# Agile Project Management Tools

Agile project management tools, such as **Jira**, **Agile Accelerator**, and **Asana**, offer **features** that support the **iterative and incremental nature** of agile development.



## BACKLOG MANAGEMENT

Agile project management tools enable **prioritization** and **organization** of **user stories, tasks, and bugs** in a **product backlog**. Teams can **add, assign, prioritize, and manage** work items.



## SPRINT PLANNING

They support the **planning of work for each sprint**, allowing teams to **assign tasks and user stories** to specific sprint cycles. This helps in **setting realistic goals** based on team capacity and sprint duration.



## BOARD VIEWS

Teams can use **Kanban/Scrum boards** to **track progress, identify bottlenecks, and manage workflow efficiently**. These boards provide a visual representation of work items in various stages of development.

# Agile Project Management Tools

Agile project management tools, such as **Jira**, **Agile Accelerator**, and **Asana**, offer **features** that support the **iterative and incremental nature** of agile development.



## COLLABORATIVE TOOLS

Features like **chat**, **forums**, **comments**, and **document sharing** to facilitate real-time communication and collaboration. They ensure alignment and easy sharing of knowledge and feedback.



## REPORTING & ANALYTICS

Teams can gain insights into **project progress**, **team performance**, and **delivery forecasts**. They can access **burndown charts** to view the **amount of work completed** and the **total work remaining**.



## INTEGRATION

These tools can be **integrated** with **version control systems** like Git, **continuous integration tools**, and other tools to streamline workflows and automate parts of the development process.

# Benefits of Agile Project Management Tools

Agile project management tools provide various benefits, including **transparency**, **flexibility**, and **real-time communication** across the development lifecycle.



## Collaboration

**Centralized communication** and **project information enhances collaboration** among distributed teams and stakeholders, ensuring that everyone is on the same page.



## Visibility

Agile tools provide **improved visibility** into every aspect of the development process, including **task status**, **sprint progress**, and the overall **project status**, facilitating **better decision-making** and **transparency** with stakeholders.



## Productivity

Streamlined workflows, automation of repetitive tasks, and clear prioritization help teams **work more efficiently**, **reducing time wasted** on low-priority tasks and manual processes.



## Quality

Continuous feedback loops, regular retrospectives, and the ability to quickly adapt to changes lead to **higher quality outcomes** and **reduced risks** associated with project delays or misaligned expectations.

# Benefits of Agile Project Management Tools

Agile project management tools provide various benefits, including **transparency**, **flexibility**, and **real-time communication** across the development lifecycle.



## Consolidation

These tools offer a **centralized platform** for storing all **project artifacts**, including project documents. This consolidation **facilitates easy access** and **version control**, ensuring everyone works with the most current information.



## Integration

Integration with testing and continuous integration tools allow developers to **see how changes affect user stories**. They can **view automated test statuses after code commits** and the **committed code's build status** on user stories.



## Scalability

Whether working on small projects or managing large-scale Salesforce implementations, agile tools can **scale to meet the needs of the team**, providing the flexibility to **adapt to changing requirements and priorities**.



## Predictability

**Clear cost estimates and timelines** offered by agile sprints, which have a known duration and scope, allow clients to **make informed decisions** regarding **feature prioritization** and **budget allocation**.

# Agile Accelerator

**Agile Accelerator** is a free agile project management tool available in the **AppExchange** marketplace that allows **managing backlog, sprints, user stories, defects, and more.**

The screenshot displays the Salesforce Agile Accelerator interface. At the top, the Salesforce logo is on the left, and a search bar and utility icons (star, plus, home, help, settings, notifications) are on the right. Below this is a navigation bar with the following items: Agile Accelerator, Work (dropdown), Work Manager (selected), Sprints (dropdown), Teams (dropdown), Kanban Board, Product Tags (dropdown), Epics (dropdown), Themes (dropdown), Reports (dropdown), Dashboards (dropdown), Projects (dropdown), Programs (dropdown), Portfolios (dropdown), and Accomplishments (dropdown).

The main section is titled 'WORK MANAGER' and 'Sample'. It includes a 'Quick search' bar, edit, filter, and refresh icons, and a 'New Wo' button. Below the header, it says '10 of 10 items - Sorted by Rank'.

| <input type="checkbox"/> | ID       | RANK ↑ | SUBJECT                                      | STATUS           | SPRINT               | ASSIGNEE       | PO |
|--------------------------|----------|--------|----------------------------------------------|------------------|----------------------|----------------|----|
| <input type="checkbox"/> | W-000002 | 1      | Point to point navigation                    | In Progress      | 2024.02a-Sample Team | Martin Gessner |    |
| <input type="checkbox"/> | W-000003 | 2      | Audio turn by turn navigation                | Triaged          |                      | Martin Gessner |    |
| <input type="checkbox"/> | W-000004 | 3      | Search by Hotel                              | Ready for Review | 2024.02a-Sample Team | Martin Gessner |    |
| <input type="checkbox"/> | W-000005 | 4      | Show traffic                                 | Fixed            | 2024.02a-Sample Team | Martin Gessner |    |
| <input type="checkbox"/> | W-000009 |        | Prompt user for location tracking permission | New              |                      |                |    |
| <input type="checkbox"/> | W-000008 |        | Maps for public transport                    | New              |                      |                |    |
| <input type="checkbox"/> | W-000007 |        | Control by voice commands                    | New              |                      |                |    |
| <input type="checkbox"/> | W-000006 |        | Text Friend ETA                              | New              | 2024.02a-Sample Team |                |    |
| <input type="checkbox"/> | W-000001 |        | No option for skipping toll roads            | New              | 2024.02a-Sample Team |                |    |
| <input type="checkbox"/> | W-000000 |        | GPS locations rendering incorrectly          | New              | 2024.02a-Sample Team |                |    |

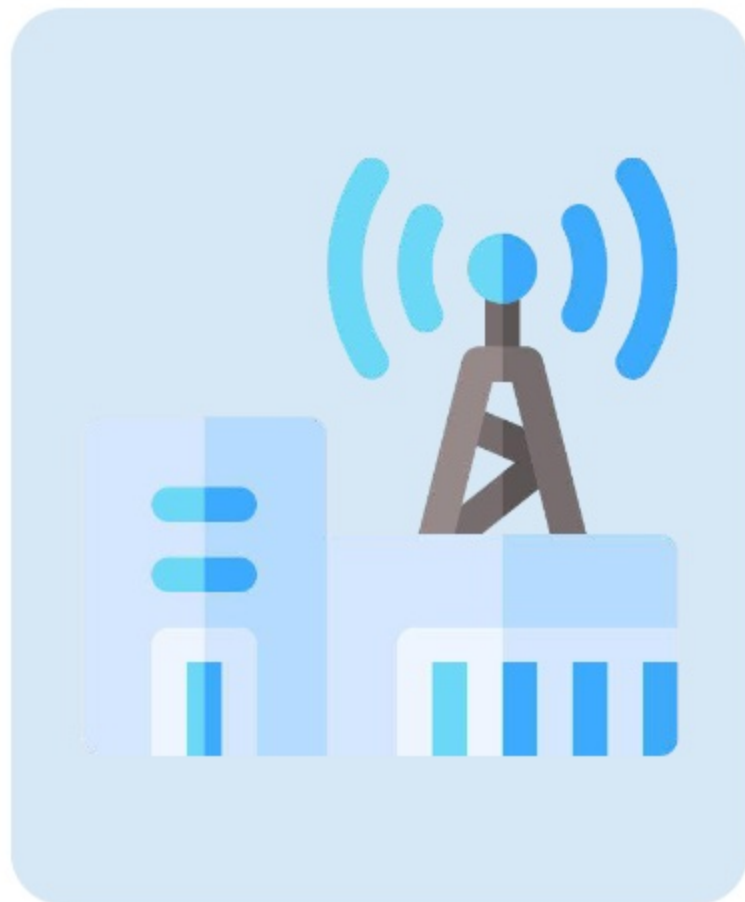
# Business Scenario

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## Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Communications, a leading provider of cloud-based solutions for the telecommunications industry, is on the verge of a major transformation. The company, known for its cutting-edge technologies and customer-centric products, is adopting Scrum to enhance its agile development processes. With teams spread across different continents, Cosmic Innovations faces the challenge of synchronizing efforts, managing workloads efficiently, and delivering high-quality software within shorter timeframes demanded by the market. The Salesforce architect of the company needs to highlight the benefits of adopting agile tools for its development processes.

# Requirements & Solutions

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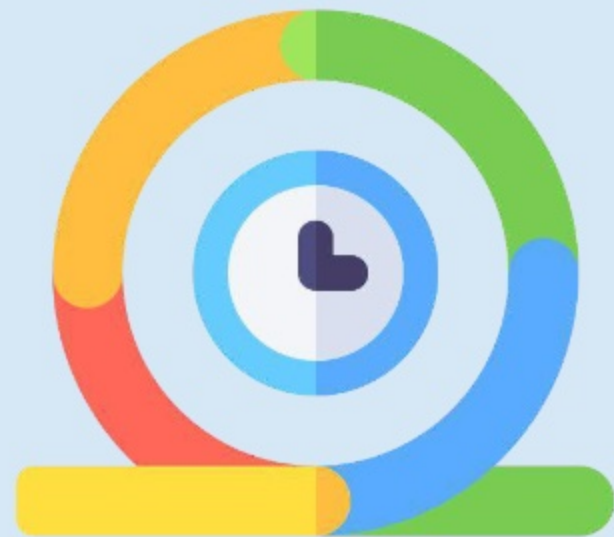
# Requirement 1



## SCENARIO

As the company adopts Scrum, the need for an agile development support tool becomes paramount. The tool must facilitate:

- ❖ Effective prioritization of the product backlog to align development efforts with strategic business goals.
- ❖ Management of sprint backlogs to ensure that each sprint is planned with achievable goals and clear priorities.
- ❖ Assignment of work, such as user stories or tasks, to individual team members, promoting accountability and transparency in the development process.



# Solution 1 (Part 1)



## SOLUTION

The company should select an **agile project management tool** that features comprehensive **product and sprint backlog management** capabilities, alongside robust **work assignment** functionalities. This tool would enable the teams to:

- ❖ **Visually organize** and **prioritize** the **product backlog**, ensuring that the most critical features are developed first.
- ❖ **Plan sprints** with a clear overview of the **sprint backlog**, allowing for better estimation and allocation of tasks.
- ❖ **Assign user stories and tasks to developers** directly within the tool, enhancing visibility and accountability across the development lifecycle.



## Solution 1 (Part 2)



### SOLUTION

- ❖ **Access burndown charts** to view the **amount of work completed** and the **total work remaining** for better sprint management.
- ❖ Use features such as **chat and comments to communicate** important information such as assignment of user stories.



## Requirement 2



### SCENARIO

Due to the growing pressure to increase development velocity while maintaining or improving quality, the company needs to:

- ❖ Consolidate project artifacts in a common repository, providing a single source of truth for all project-related information.
- ❖ Integrate the agile issue tracker with application testing and continuous integration (CI) tools.



## Solution 2



### SOLUTION

Most agile project management tools offer a **centralized platform** for the **consolidation of all the project artifacts**, improving access to information and facilitating better decision-making. Integrating the company's agile tool with **testing and continuous integration (CI) tools** would enable developers to see **automated test statuses** and the **build status of committed code** directly on the **user stories**, streamlining the development and QA processes. This visibility would assist in identifying and resolving issues more quickly and enhance collaboration between the development and QA teams.



## Learn More

-  [How Salesforce Developer Experience Changes the Way You Work](#)
-  [Set Up Salesforce DX Tools](#)
-  [Learn Salesforce Agile Practices](#)
-  [Agile Accelerator](#)

